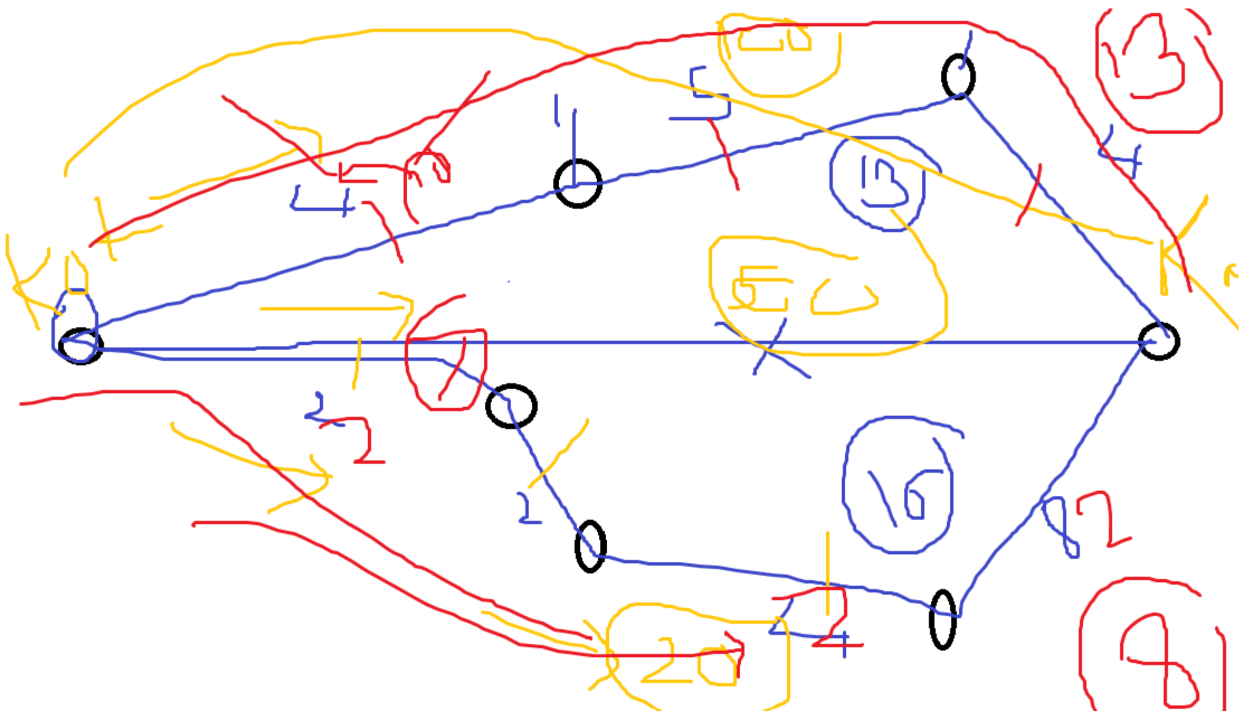
DIJKSTRA'S ALGORITHM (10m)

Definition:(what,why,when,where…..) 1m

Dijkstra's Algorithm is a *greedy algorithm* used to find the shortest path from a single source vertex to all other vertices in a weighted, non-negative edge graph.

Works for directed and undirected graphs with non-negative edge weights.

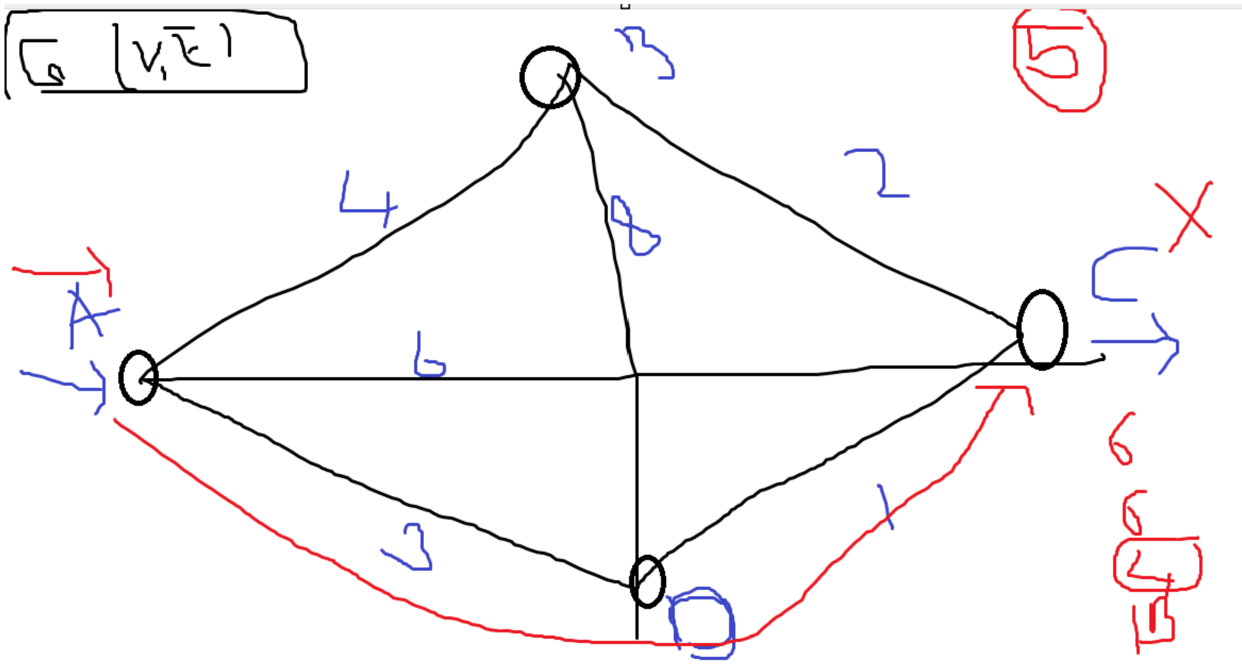
A Greedy Algorithm is a problem-solving approach where decisions are made step-by-step by choosing the option that looks best at the moment (locally optimal), with the hope that this leads to a global optimum.



Formula/ Terminology: (1m)

Graph G(V, E): Set of vertices V, and edges E.

Weight (u, v): Cost or distance to go from vertex u to vertex v.



Proof: (1m)

GIVEN / INITIALIZE: (ASSUMPTION)

Step 1:Initialize (Depend on the graph)

-vertices (A,B,C,D)

-edges (AB=4 , AC,AD,BC,BD,ABCD,ACBD…….

Starting vertex/node A…..

Ending node /Destination D….

Step2 :Path/Visit vertices/route……

Start: A

End: D

Path: A-B-D ,A-D ,A-C-D ,A-B-C-D ,A-C-B-D

Step: find the shortest/minimum that is the answer

LIMITATION

Doesn't work for negative edge weights.
(Use Bellman-Ford Algorithm if negative weights are involved.)

ASSIGNMENT:

2Diffrent graphs (N number of connected vertices with weighted edges)

Set a start vertex and end vertex

Find the shortest path (dijkstras algorithm)